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Superseding
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28 February 1964
with Amendment 1 of
22 January 1965

MILITARY SPECIFICATION

CUTTER, DELAY, PROPELLANT ACTUATED:
M21 (2 SEC) AND M22 (10 SEC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Cutters, Delay, Propellant Actuated M21 (2 sec delay) and M22 (10 sec delay).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons, General Specification for
MIL-W-5625	Webbing, Nylon, Tubular
MIL-S-13572	Springs, Helical, Compression And Extension
MIL-C-13739	Composition, Delay
MIL-P-20444	Primer, Percussion, M42, Parts for and Loading, Assembling and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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MIL-P-22264	Powder, Ignition, Gasless, A-1A
MIL-P-25732	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 275°F
MIL-W-83420	Wire Rope, Flexible, For Aircraft Control

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-413	Visual Inspection Guide for Rubber o-rings
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods
MIL-STD-1168	Ammunition Lot Numbering
MIL-STD-45662	Calibration System Requirements

DRAWINGS

NAVORDSTA, Indian Head (Code Ident 14083)

DL8875978	Cutter, Delay, Propellant Actuated, M21
DL8875979	Cutter, Delay, Propellant Actuated, M22
8863435	Box, Fiberboard (Inner) Packing Ammunition for Cutter, Delay, M21
8863436	Box, Packing Ammunition for Cutter, Delay, M21
8863437	Box, Fiberboard (Inner) Packing Ammunition for Cutter, Delay, M22
8863438	Box, Packing Ammuniton For Cutter, Delay M22
8873322	Cutter Tester

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8875982	Firing Pin Assembly
8875988	Spring
9244056	Box, Fiberboard (Outer) Alternative Packing, Ammunition for Cutter, Delay, M22
9251561	Box, Fiberboard (Outer) Alternative Packing, Ammunition for Cutter, Delay, M21

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

WR-43A	Classification of Characteristics

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108-73	Steel Bars, Carbon, Cold Finished Standard Quality
ASTM A228-71	Steel Wire, Music Spring Quality
ASTM A276-75	Stainless and Heat-Resisting Steel Bars and Shapes
ASTM B211-75	Aluminum-Alloy Bars, Rods, and Wire
ASTM B221-76	Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tube

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association, specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 General. The cutters shall comply with all requirements specified on drawings listed on DL8875978 or DL8875979 and with all requirements specified in applicable specifications (see 4.8.1).

3.2 First article. When specified (see 6.2) the contractor shall furnish sample units for first article inspection and approval (see 4.6 and 6.2).

3.3 Materials. All materials used in the manufacture of the cutters shall conform strictly to the specifications referenced on the respective drawings (see 4.3) unless specified approval in writing covering a departure therefrom has been obtained from the procuring activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. Physical properties required shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering (see 4.8.2).

3.4 Primer X-ray. All primers shall be x-rayed and screened prior to assembly into the primer retainer (see 4.8.4).

3.5 Primer sensitivity. The primer shall fire when drop tested at an energy level of 26 inch-ounces in a primer retainer assembly (see 4.8.5).

3.6 Spring load and spring embrittlement. The springs shall not break and shall meet the requirements of drawing 8875983 when tested as specified in 4.8.6 and 4.8.7.

3.7 Corrosion resistance. All component parts (with the exception of the delay body, washer (retainer) and the primer retainer) shall exhibit no evidence of white corrosion products after being subjected to 100 hours of salt spray testing. The above cited components shall exhibit no evidence of white corrosion products after being subjected to 24 hours of salt spray testing (see 4.8.8).

3.8 Airtightness. All knife delay assemblies shall be leak tested and shall not exhibit a leakage rate in excess of $1x10^{-4}$ cc of gas or air per second at a pressure differential of one atmosphere (see 4.8.9).

3.9 Waterproofness of knife - delay assembly. The knife delay element shall show no evidence of leakage when tested as specified in 4.8.10.

3.10 Lanyard assembly. The lanyard assembly shall be able to withstand a tensile load of 200 lbs minimum without any separation of parts (see 4.8.11).

3.11 Radiographic inspection. After completion of assembly, all cutters shall be x-rayed for proper assembly and presence of components. Any evidence of misassembly or missing parts shall be cause for rejection of that assembly (see 4.8.12).

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3.12 Staking. The staking between the firing pin assembly and case shall withstand a torque of 25 in-lbs without any turning or loosening of the components (see 4.8.13).

3.13 Vibration. The cutter shall withstand the vibration test without evidence of any disintegration of the primer, delay column, propellant or derangement of any component (see 4.8.14)

3.14 Nonwithdrawal and withdrawal of cotter pin. The safety cotter pin shall not withdraw from the cutter when a pull force of 10 lbs minimum is applied to the pin. The pin shall withdraw when a pull force of 30 lbs maximum is applied to the cotter pin (see 4.8.15).

3.15 Firing pin release and non-release. The firing pin shall not be released when a force of 10 lbs minimum is gradually applied to the lanyard assembly and the firing pin shall be released when a force equivalent to dropping a 20 lb weight from a height of 3 ± 1/3 inches is applied to the lanyard assembly (see 4.8.16).

3.16 Function.

3.16.1 Functioning over temperature range. The cutter shall function satisfactorily without mechanical failure (see 6.3.1) throughout the temperature range of -65°F to +165°F by completely severing two (2) reefing lines within the specified time in Table I when tested according to 4.8.17.

TABLE I. Functioning delay times (see 6.3.2).

Temperature	Time Delay M21 (seconds)	Time Delay M22 (seconds)
-65°F	3.0 (Maximum)	13.0 (Maximum)
+70°F	2.0 ± 0.4	10.0 ± 1.0
+165°F	1.2 (Minimum)	8.0 (Minimum)

3.16.2 Functioning at high set-back. The cutter shall function satisfactorily as evidenced by high order functioning of the propellant charge when a shock force of 700 G's minimum is applied to the cutter (see 4.8.18).

3.17 Packaging, packing and marking. The packaging and packing of cutters and the container marking shall be as specified in Section 5.

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3.18 Workmanship. The workmanship requirements are as specified on the drawings, specifications and the following. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, butts, deformations and missing operations which may affect assembly, serviceability, functioning, operation, appearance or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer. Surface coatings shall be continuous except for a few light scratches not exposing base material.

3.18.1 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and insepction facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.2 Inspection. Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

A. Materials inspection (See 4.3)

B. First article inspection (See 4.6)

C. Quality conformance inspection (See 4.7)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in Table II, used in fabricating the cutters, are in accordance with the applicable referenced specification or requirements prior to such fabrication.

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TABLE II. Materials inspection.

Material	Requirement Paragraph	Applicable Specifications
Aluminum	3.3	ASTM B211 ASTM B221
Ignition Composition	3.3	MIL-P-22264
O-Rings	3.3	MIL-P-25732
Spring Steel Wire	3.3	MIL-S-13572 ASTM A228
Steel	3.3	ASTM A108 ASTM A276
Delay Composition	3.3	MIL-C-13739
Primer	3.3	MIL-P-20444
Steel Cable	3.3	MIL-W-83420

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-810.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C) and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified as minor.

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4.6 First article inspection. First article inspection shall be performed after award of contract and prior to production. After inspection and provisional acceptance at the Source, sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract. First article inspection shall be performed on sample units which have been manufactured in the same manner using the same materials, equipment, processes and procedures as to be used in the regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.6.1 Sample size. Unless otherwise specified, the first article sample shall consist of 28 assembled units and 10 complete sets of components and subassemblies (i.e. 10 each of every component and 10 each of the lanyard and firing pin assemblies and 10 knife delay assemblies) and one set of metal components without plating or protective finish for dimensional and visual examination and tests.

4.6.2 Inspection routine. The first article sample shall be subjected to the inspections specified in Table III.

4.6.3 First article sample failure. The first article sample must meet the requirements referenced in Table III. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any or all of the stated requirements. In the event of failure of any sample, the Government reserves the right to require the contractor to take corrective action and submit a new sample or portion thereof. Until a First Article Inspection sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the procuring activity.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A inspection and Group D inspection. Group A and Group D inspections shall be done prior to delivery.

4.7.1.1 Submission of product. Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provision of MIL-STD-105.

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TABLE III. First article inspection.

Inspection	Requirement Paragraph	Test Method Paragraph	Number of Samples to be Inspected
Visual and Dimensional Inspection of Components	3.1	4.8.1	10 Components
Hardness	3.3	4.8.2	5 Components
Spring Load	3.6	4.8.6	5
Embrittlement of Spring	3.6	4.8.7	5
Corrosion Resistance	3.7	4.8.8	3 Components
Airtightness	3.8	4.8.9	10
Waterproofness of Knife Delay Assembly	3.9	4.8.10	10
Lanyard Assembly Pull Test	3.10	4.8.11	10
Radiographic Inspection	3.11	4.8.12	28
Staking	3.12	4.8.13	28
Vibration	3.13	4.8.14	21
Non-withdrawal and Withdrawal of Cotter Pin	3.14	4.8.15	28
Firing Pin Release and Non-Release	3.15	4.8.16	28
Functioning over Temperature Range	3.16.1	4.8.17	21
Functioning at High Set Back	3.16.2	4.8.18	7

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4.7.1.2 Lot size. A lot shall consist of an identifiable group of assembled cutters, not more than 1200, consisting of metal parts produced by a homogeneous process and containing:

A. Primers from not more than one lot.

B. Ignition composition from not more than one lot.

C. Delay composition from not more than one lot.

D. Propellant from not more than one lot.

4.7.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in Table VI in the order shown.

4.7.1.3.1 Sampling plan. Examination for critical characteristics shall be done on a sample size of 200. For lots of 200 or under, 100% inspections for all critical characteristics shall be done. Examination for major or minor characteristics shall be performed on a defect (see 6.3.3) basis in accordance with the classification on each applicable drawing of DL8875978 or DL8875979 or as contained herein, using the sampling plan and criteria in Tables IV and V. For an individual defect (single characteristic), the criteria in Table IV shall be used. For total defects within a major or minor group, the criteria in Table V shall be used. In addition to the characteristics shown on assembly drawing 8875978 (for the M21 cutter) or assembly drawing 8875979 (for the M22 cutter), examination for characteristics listed on Table VII shall also be made. All non-conforming material shall be rejected.

TABLE IV. Sampling plan, individual defects.

		Major		Minor	
Lot Size	Sample Size	Accept	Reject	Accept	Reject
1-280	50 1/	0	1	2	3
281-1200	80	1	2	3	4
1201-3200	125	2	3	5	6
3201-10000	200	3	4	7	8

1/ If sample size equals or exceeds lot size do 100 per cent inspection. When 100 per cent inspection is done all nonconforming material is rejected, conforming material accepted.

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TABLE V. Sampling plan, total defects.

Number of Characteristics in Class	Sample Size for Major and Minor Defects				
	50	80	125	200	315
	Total allowable major defects per sample size				
2 - 3	-	1	2	3	4
4 - 7	1	2	3	4	5
8 - 19	2	3	4	5	7
20 - 43	3	4	5	8	10
Over 43	4	6	8	11	14
	Total allowable minor defects per sample size				
2 - 3	1	2	3	4	5
4 - 7	2	3	4	5	7
8 - 16	3	4	5	8	10
17 - 48	4	6	8	11	14
Over 48	6	8	11	17	20

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TABLE VI. Group A inspection.

Inspection	Requirement Paragraph	Test Method Paragraph	Quantity Tested	Accept/Reject Criteria
Visual and Dimensional Examination	3.1	4.8.1	Sample per Tables IV and V	4.7.1.3.2
Protective Finish	3.3	4.8.3	Per Spec.	Per Spec.
Primer X-Ray	3.4	4.8.4	All	No Failures
Primer Sensitivity	3.5	4.8.5	See 4.7.1.3.4	No Failures
Spring Load	3.6	4.8.6	10	No Failures
Embrittlement of Springs	3.6	4.8.7	10	No Failures
Airtightness	3.8	4.8.9	All	4.7.1.3.2
Waterproofness of Knife Assembly	3.9	4.8.10	65	4.7.1.3.2
Lanyard Assembly Pull Test	3.10	4.8.11	All	4.7.1.3.2
Radiographic Inspection of Assembled Cutters	3.11	4.8.12	All	4.7.1.3.2
Withdrawal and Non-Withdrawal of Cotter Pin	3.14	4.8.15	25	4.7.1.3.2
Release and Non Release of Firing Pin	3.15	4.8.16	All	4.7.1.3.2

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TABLE VII. Classification of assembly defects.

Classification	Defect	Method of Inspection
CRITICAL		
1.	All parts are present and are assembled correctly	Visual/X-Ray
MAJOR		
101.	Assembly damaged	Visual
102.	Stake inadequate	Visual
103.	Marking, clear and legible	Visual
MINOR		
None defined		

4.7.1.3.2 Component or assembly failures. Any components or assembled cutters failing to meet requirements shall be cause for rejection of that item. In the waterproofness of knife delay assembly test, if two failures occur, the lot is rejected. If only one failure is found, a second sample of 130 assemblies shall be tested. When a total of 2 defectives are encountered in the combined first and second sample testing will be stopped and the lot shall be rejected. In the non-withdrawal and withdrawal of safety cotter pin test, if more than one failure occurs the lot shall be rejected. If only one defective is found, a second sample of 50 shall be tested. If one or more defectives are found in the second sample, the lot shall be rejected. In visual and dimensional examinations, defects exceeding the number specified in Tables IV and V shall result in lot rejection.

4.7.1.3.3 Rejected lots. If an inspection lot is rejected, disposition of the lot will be as directed by the procuring activity. In the event the contractor is directed to rework the lot, a written procedure for the rework will be submitted to the responsible Government engineering activity for approval before commencing the rework. Such lots shall be separate from new lots and shall be clearly identified as reworked lots. in accordance with MIL-STD-1168.

4.7.1.3.4 Primer sensitivity sample. The sample for primer sensitivity shall be in accordance with Table VIII.

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TABLE VIII. Primer sensitivity sample.

Lot Size	Primer Sensitivity Sample Size
3 to 40	5
41 to 110	10
111 to 800	35
801 and over	50

4.7.1.4 Group D inspection. Group D inspection shall consist of the tests specified in Table IX and shall be made on sample units which have undergone and passed the Group A inspection. The sample shall consist of thirty (30) cutter assemblies randomly selected from each lot unless otherwise specified by the procuring activity. An additional quantity of 10 assemblies for functional test at high set back is required from those lots from contracts not requiring first article inspection (see 6.2).

4.7.1.4.1 Test failure. The cutter assembly shall be classed defective if it releases the firing pin under minimum force, fails to fire under maximum force, fails to meet the specified time requirement, or fails to completely sever both reefing lines. Failure of any unit to comply with the requirements of 3.13, 3.15, or 3.16 shall be cause for lot rejection.

4.7.1.4.2 Retest. There shall be no retests of a failed lot unless directed by the contracting officer. The retest shall be done under direction of the responsible Government engineering activity.

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TABLE IX. Group D inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	QUANTITY TESTED	ACCEPT/REJECT CRITERIA
Vibration 1/	3.13	4.8.14	21	4.7.1.4.1
Release and Non-Release of Firing Pin	3.15	4.8.16.1	30	4.7.1.4.1
Functional Test Over Temperature Range	3.16.1	4.8.17	30	4.7.1.4.1
Functional Test at High Set Back 1/	3.16.2	4.8.18	10	4.7.1.4.1

1/ To be done only for those contracts for which a first article inspection was not done (see 6.2).

4.7.2 Packaging inspection. Inspection of packaging and packing components shall be made to determine that they conform to requirements of Drawings 8863435, 8863436 and 9251561 for M21 cutter and requirements of Drawings 8863437, 8863438 and 9244056 for M22 cutter. Acceptance or rejection will be made using classification of defects in Table X and criteria in Tables IV and V. All non-conforming packaging and packing components shall be rejected.

4.7.3 Quick leak test of sealed barrier bag. A sample shall be selected from each lot for this test in accordance with S4 of Table I of MIL-STD-105. Testing shall be as referenced on Drawing 8863435 for M21 cutters or 8863437 for M22 cutters. Failure of one or more units of the sample in meeting the requirements shall reject the lot subject to 100 percent testing.

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TABLE X. Classification of packaging defects.

CLASSIFICATION	DEFECT	METHOD OF INSPECTION
A. Box, Fiberboard (Inner) Packing Ammunition, Drawing 8863435 (M21 Cutter) or Drawing 8863437 (M22 Cutter)		
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
	201. Contents loose	Manual
	202. Box broken or damaged	Visual
	203. Supports and fillers missing or improperly installed	Visual
	204. Sealing strip missing or improperly applied	Visual
	205. Cut, damaged or improperly sealed bag	Visual
B. Box, Packing, Drawing 8863436 (M21 Cutter) or Drawing 8863438 (M22 Cutter)		
CRITICAL		
	1. Explosive marking not present or illegible	Visual
MAJOR		
	101. Exceeds weight maximum	Visual/Manual
MINOR		
	201. Contents loose	Manual
	202. Strapping missing, loose, broken or improperly assembled	Visual/Manual
	203. Contents exposed	Visual
	204. Box damaged	Visual
	205. Number of units incorrect	Visual

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C. Box, Fiberboard (Outer) Alternative Ammunition Packing, Drawing 9251561 (M21 Cutter) or Drawing 9244056 (M22 Cutter)

CRITICAL		
1.	Explosive marking not present or illegible	Visual
MAJOR		
101.	Exceeds weight maximum	Visual/Manual
MINOR		
201.	Contents loose	Manual
202.	Box broken or damaged	Visual
203.	Number of units contained incorrect	Visual
204.	Tape incomplete	Visual

4.8 *Tests and examinations.*

4.8.1 Visual and dimensional examinations. Visual examinations shall be made for such defects as defective metal, burr, sharp corners, foreign matter, defective protective finish and/or marking missing, incorrect or illegible, thread damaged or not full, holes, slots, radii or chamfers missing. Four power (4X) magnification may be used in performing visual examination for such defects. An appropriate gage or measuring device will be used to check all other characteristics shown on the drawing. Inspection for rubber o-rings shall be in accordance with MIL-STD-413 and MIL-A-2550.

4.8.2 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine the hardness values specified on the applicable drawing.

4.8.3 Protective finish. Test procedures for protective finish shall be in accordance with the component drawing.

4.8.4 Primer X-Ray (C 1). Prior to assembly into the primer retainer the primer shall be x-rayed with the side of cup perpendicular to the rays. The X-ray negatives shall be identified by item nomenclature and lot number. After screening and indicating defective primers the X-ray photos shall be forwarded to the responsible Government engineering activity (see 6.2).

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4.8.5 Primer sensitivity. Test shall be made on primers assembled into primer retainers. The primer shall be crimped and sealed in the retainer in accordance with the assembly drawing. Drop test at 13 inches (adjusted) using a stainless steel ball weighing 1.94 ± .02 ounces in a primer sensitivity tester with a primer retainer holder approved by the procuring activity. The drop height of 13 inches shall be adjusted taking into consideration the actual weight of the ball so that an energy level of 26 inch-ounces shall be achieved. All height measurements shall be made between the head of the firing pin and the bottom of the suspended steel ball. The primer sensitivity test equipment shall be checked to insure a center impaction contact between the dropped steel ball and head of firing pin.

4.8.6 Spring load. The spring shall be assembled into the test equipment and force applied until the spring is compressed to the larger height specified in Drawing 8875988. The applied force shall be recorded. Force shall be re-applied until the unit is compressed to the smaller height specified in Drawing 8875988 and the applied force recorded. Any spring exhibiting a compression force at either height, outside the load range permitted by Drawing 8875988 shall be classified defective.

4.8.7 Spring embrittlement. The spring shall be held at each end and stretched until the minimum point of permanent distortion is reached. If the spring breaks before reaching that point, it shall be classed defective. Springs so tested shall be discarded.

4.8.8 Corrosion resistance. Testing shall be performed as specified in method 811.1 of FED-STD-151.

4.8.9 Airtightness. Each knife delay assembly shall be tested in a dry gas or dry air leak tester.

4.8.10 Waterproofness of knife assembly. The knife delay assembly shall be placed under water in a vacuum chamber. The water shall contain the wetting agent Naccon NR (Sodium alkyl aryl sulfonate) or equal. The chamber shall be evacuated to a minimum pressure differential of 5 psi. Any assembly emitting bubbles, at any frequency, from one or more points shall be classed defective and discarded. After subjection to test, the assembly shall be rinsed thoroughly in fresh running water to remove the wetting agent and dried at room temperature.

4.8.11 Lanyard assembly pull test. The lanyard assembly shall be placed in the applicable test equipment wherein it shall be held at the firing pin release and through the lanyard eye. The specified load shall be applied. No dwell time is required. Any unit which separates shall be classified defective

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4.8.12 Radiographic examination. Each assembled cutter shall be radiographically examined in accordance with MIL-STD-453, with the exception that a fully assembled cutter will be substituted for the penetrameter. The cutters shall be positioned for the most revealing exposure with the long axis of the cutter perpendicular to the rays of the x-ray machine. Examine all assemblies by radiographic plates and any observable imperfections in components or assembly shall be cause for cutter rejections. All assemblies shall be identified with serial numbers prior to x-ray examination. The cutter serial numbers shall be in consecutive order beginning with the number (001) in each production lot. The assemblies for radiographic inspection shall be arranged on trays or boards in consecutive numerical order and each radiographic plate shall carry a permanent identification of the assemblies displayed thereon. Any missing serial numbers are to be identified on the x-ray. The radiographic plate identification shall include the complete lot number, as marked on the cutter, and the span of serial numbers displayed. Radiographic plates of the entire production lot shall be forwarded to the activity conducting the ballistic acceptance testing. Film interpretation of the radiographic plates shall verify that the final assembly is properly assembled and includes all components. Defective assemblies found during x-ray review are to be marked on the x-ray prior to shipment and those defective items shall be removed from the production lot.

4.8.13 Staking (M 102). The assembled cutter shall be placed in a holding device in which the prescribed amount of torque can be applied to the retainer to verify that the staking between case and retainer. meets 3.12.

4.8.14 Vibration. The selected sample will undergo transportation vibration test as specified by procedure II in Test 401 of MIL-STD-331.

4.8.15 Non-withdrawal and withdrawal of cotter pin. The assembled cutter shall be firmly mounted in a holding fixture. A force equal to the minimum specified withdrawal force shall be applied, without impact, to the cotter pin so as to tend to remove the pin from the assembly. The force shall act in a direction parallel to the length of the pin and shall be applied at the eye. If the safety cotter pin is not withdrawn when the minimum force is applied, the maximum force shall be applied in the same manner as specified above. Any cotter pin which is withdrawn when the minimum force is applied or is not withdrawn when the maximum force is applied shall cause the assembly to be classed as defective.

4.8.16 Firing pin release and non-release. The firing pin assembly (Dwg. 8875982) shall be secured in a fixture. A pull force shall be applied to the lanyard in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

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4.8.16.1 Firing pin release in functional test. In conjunction with functioning over temperature range tests, the force required to release the firing pin shall be recorded on the ballistic firing record.

4.8.17 Functioning over temperature range. The cutter shall be mounted in test fixture, Drawing 8873322. Two (2) reefing lines (1,000 pound tensile strength, ½ inch, MIL-W-5625) shall be passed through the hole in the cutter. One end of the reefing lines shall be anchored and a 50 lbs load attached to the free ends of the reefing lines. The lanyard shall then be pulled to release the firing pin. A ballistic trace which records the force applied to the lanyard to release the firing pin and the time delay (see 6.3.2) shall be made for each firing.

4.8.17.1 Temperatures. The sample for functioning over the temperature range shall be divided into three equal groups and temperature conditioned as shown in Table XI. The units shall be tested within ten (10) minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be re-conditioned for a minimum of one hour.

NOTE: The first article sample and 21 of the group D inspection samples from contracts not requiring first article inspection shall be vibrated per 4.8.14 before temperature conditioning.

4.8.17.2 Temperature conditioning. An inert cutter containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be one (1) hour longer than the time necessary to bring the element contained in the dummy cutter to the required temperature.

4.8.18 Functioning at high set back. The cutter assembled to the test fixture (Drawing 8873322) shall be initiated by release of the firing pin. Midway thru the delay burning time the specified shock shall be applied to the cutter in the direction of burning of the delay. Observation shall be made for complete functioning as evidenced by high order functioning of the propellant charge. This test is performed at ambient temperature.

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TABLE XI. Conditioning requirements for "Funtioning" test.

Temperature	First Article Inspection Sample	Group D Sample	Group D Sample (for lots without first article inspection)	
	Vibrated	Non-Vibrated	Vibrated	Non-Vibrated
-65° to -70°F	7	10	7	3
65° to 75°F	7	10	7	3
160° to 165°F	7	10	7	3

5. PACKAGING

5.1 Preservation-packaging. Preservation packaging shall be in accordance with requirements of Drawing 8863435 for M21 cutters and of Drawing 8863437 for M22 cutters.

5.2 Packing. Packing level shall be as specified by the procuring activity (see 6.2);

	For M21 Cutters	For M22 Cutters
Level A - In accordance with	Drawing 8863436	Drawing 8863438
Level B - In accordance with	Drawing 9251561	Drawing 9244056

5.3 Marking (M103). Marking shall be in accordance with MIL-STD-129 and requirements of Drawings 8863435, 8863436 and 9251561 for M21 cutters or Drawings 8863437, 8863438 and 9244056 for M22 cutters.

6. NOTES

6.1 Intended use. The loaded assembly is used to sever one or two ½ inch reefing lines and is a component part of an aerial delivery system for cargo and drone recovery.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, Number and date of this specification

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b. Whether First Article Sample is required, and if so, specify the test activity

c. Production lot size (see 4.7.1.2)

d. Items of data required for materials inspection and production lot (see 6.2.2)

e. Government facility designated to perform ballistic tests

f. Ballistic test sample size if other than 30 units (see 4.7.1.4)

g. Activity to receive radiographs of primers before assembly and assembled cutters (see 4.8.4 and 4.8.12)

h. Level of packing required

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	Certificate of Compliance	DI-E-2121
4.8.4 & 4.8.12	Radiographs	-
6.2.2.1	Data Cards	-

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.2.2.1 Data cards. The contract should state the format, quantity and destination of the data cards.

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Time delay. Time delay is defined as the time elapsed from indication of firing pin release on the ballistic record until severing of the reefing lines.

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6.3.3 Defect. A defect is any non-conformance of a characteristic with specified requirements.

6.3.4 Group A inspection. Nondestructive inspections of all items produced or all samples from an inspection lot to demonstrate product compliance with contractual requirements. Group A inspections examine characteristics most affected by variations in production processes or skills and functions vital to successful completion of the design mission.

6.3.5 Group D inspection. Destructive tests that consume all of design service life. Articles tested in Group D inspections are considered completely expended and will not be issued for service use. Tests are performed on few samples based on production quantities.

6.3.6 AR. As required (when shown on drawings or parts lists).

6.4 Quality assurance terms and definitions. Reference should be made to MIL-STD-109 to define quality assurance terms used.

6.5 Explosive hazard classification information.

Explosive hazard classification (Quantity-Distance) - Class 1

Storage Compatibility Grouping - Group B, E, or N

Department of Transportation (DOT) Hazard - Class C

DOT Marking - "Explosive Power Device - Class C - Handle Carefully - Keep Fire Away"

6.6 Safety precautions. The safety precautions requirements of the "Contractors Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Defense Acquisition Regulation (DAR) 1-323. When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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